

Section 2 Cheat Sheet -- Planning Compute, Storage & Network

GCP Study Notes

Study Notes April 2026

COMPUTE – Decision Flowchart

Need full OS / GPU / Windows / lift-and-shift?

--> Compute Engine

Need container orchestration / K8s?

--> GKE (Autopilot for less ops, Standard

Stateless container, no K8s overhead?

--> Cloud Run

Simple event-driven function?

--> Cloud Functions (Gen 2)

Not sure?

--> Cloud Run (safest default)

Compute Quick-Compare

	Compute Engine	GKE	Cloud Run	Cloud Functions
Model	IaaS (VMs)	CaaS (K8s)	Serverless containers	FaaS
State	Stateful	Both	Stateless	Stateless
Scale to zero	No	No	Yes	Yes
Startup	Minutes	Seconds	Seconds	Seconds
Max timeout	Unlimited	Unlimited	60 min	60 min (Gen2)
Pricing	Per-second (min 1 min)	Cluster fee + VMs	Per request + CPU/mem	Per invocation + compute

Compute Engine Machine Types

- Naming: FAMILY-RATIO-vCPUS (e.g. e2-standard-4)
- Ratios: standard = 4 GB/vCPU, highmem = 8 GB, highcpu = 1 GB
- Custom types: N1, N2, N2D, E2 only – memory in 256 MB increments

Family	Prefix	Use
General-purpose	E2, N2, N2D, C3	Balanced workloads
Compute-optimized	C2, C2D, H3	HPC, batch, gaming
Memory-optimized	M1, M2, M3	SAP HANA, in-memory DB
Accelerator-optimized	A2, A3, G2	ML training, GPU

Spot VMs

- **60-91% discount**, can be reclaimed with 30s notice
- Spot = no max lifetime; Preemptible (legacy) = 24h max
- Good for: batch, CI/CD, rendering, ML training with checkpointing
- Bad for: production serving, databases, anything needing SLA

Compute Pricing Levers

Mechanism	Discount	Commitment
On-demand	0%	None
Sustained Use (auto)	Up to 30%	Run >25% of month
CUD 1-year	Up to 57%	1-year commitment
CUD 3-year	Up to 70%	3-year commitment
Spot VMs	60-91%	None (can be reclaimed)

- Free tier: 1x e2-micro/month (us-west1, us-central1, us-east1 only)

GKE Modes

	Standard	Autopilot
Node mgmt	You	Google
Pricing	\$0.10/hr cluster + VMs	Per pod resources
Best for	Full control	Most workloads

Cloud Run vs Cloud Functions

	Cloud Run	Cloud Functions
Deploy unit	Container image	Source code
Languages	Any	Limited runtimes
Concurrency	Multiple req/instance	1 (Gen1) / many (Gen2)
Timeout	60 min	9 min (Gen1) / 60 min (Gen2)

STORAGE & DATABASES – Decision Flowchart

Relational + single region + <64 TB?	--> Cloud SQL
Relational + PostgreSQL + need HTAP perf?	--> AlloyDB
Relational + global + strong consistency?	--> Cloud Spanner
Analytical / data warehouse / SQL on PBs?	--> BigQuery
NoSQL document / mobile+web / real-time sync?	--> Firestore (Native mode)
NoSQL wide-column / time-series / >1 TB / IoT?	--> Bigtable
Object storage (files, backups, media)?	--> Cloud Storage
Block storage for VMs?	--> Persistent Disk
Shared filesystem (NFS)?	--> Filestore

Database Quick-Compare

	Cloud SQL	AlloyDB	Spanner	BigQuery	Firestore	Bigtable
Type	Relational	Relational (PG)	Relational	Analytical	Document NoSQL	Wide-column NoSQL
Workload	OLTP	OLTP+OLAP	OLTP	OLAP	Mobile/Web	Time-series, IoT
Scale	Vertical	Vertical	Horizontal	Serverless	Serverless	Horizontal
Max size	64 TB	128+ TB	Unlimited	Unlimited	1 TB/db	Unlimited
Consistency	Strong	Strong	Strong (global)	Eventual	Strong	Eventual
SLA	99.95-99.99%	99.99%	99.999%	99.99%	99.999%	99.99%
Cost entry	Low	Medium	High	Free tier	Free tier	Medium

Cloud Storage Classes

Class	Min Duration	Retrieval Cost	Use Case
Standard	None	Free	Hot data, frequently accessed
Nearline	30 days	\$	< 1x/month access
Coldline	90 days	< 1x/quarter access Archive 365 days \$	< 1x/year, compliance

- Storage cost: Standard > Nearline > Coldline > Archive
- Retrieval cost: Archive > Coldline > Nearline > Standard (free)
- All classes have **same millisecond latency** (not like AWS Glacier!)
- Early deletion = charged for remaining min duration
- Lifecycle rules can auto-transition objects between classes

Persistent Disk Types

Type	Performance	Use
pd-standard	Low (HDD)	Bulk, sequential I/O
pd-balanced	Medium (SSD)	General purpose (default)
pd-ssd	High (SSD)	Databases, random I/O
pd-extreme	Highest (SSD)	Mission-critical DBs

- **Zonal PD:** single zone, standard cost
- **Regional PD:** 2 zones, ~2x cost, faster failover
- **Local SSD:** ephemeral, extreme IOPS, data lost on stop/delete

NETWORK – Decision Flowchart

Load Balancer Selection

EXTERNAL traffic:

HTTP/S? --> External Application LB (global, L7)

TCP + SSL term? --> External Proxy Network LB (global, L4)

TCP/UDP raw? --> External Passthrough Network LB (regional, L4)

INTERNAL traffic:

HTTP/S? --> Internal Application LB (regional, L7)

TCP/UDP? --> Internal Passthrough Network LB (regional, L4)

Load Balancer Quick-Compare

LB	Scope	Layer	Protocol	Key Feature
Ext Application	Global	L7	HTTP/S	URL maps, CDN, Cloud Armor, anycast IP
Int Application	Regional	L7	HTTP/S	Internal microservices routing
Ext Proxy Network	Global	L4	TCP/SSL	SSL termination, non-HTTP
Ext Passthrough Network	Regional	L4	TCP/UDP	Preserves source IP, UDP support
Int Passthrough Network	Regional	L4	TCP/UDP	DB HA, internal services

Network Service Tiers

	Premium (default)	Standard
Routing	Google private backbone	Public internet
Latency	Lower	Higher
Global LB	Yes	No (regional only)
Cost	Higher	Lower
Use	Production	Dev/test, cost-sensitive

VPC Essentials

- VPC = **global**, Subnet = **regional**, VM = **zonal**
- Auto-mode VPC: subnets created automatically in all regions
- Custom-mode VPC: you define subnets manually

Resource Scope

Global	Regional	Zonal
VPC, firewall rules, routes	Subnets, regional PD, Cloud NAT	VMs, zonal PD, local SSDs
External IPs, images, snapshots	Cloud Router, managed instance groups	GKE node pools
Some load balancers, CDN	Internal IPs	

Connectivity to On-Premises

Option	Bandwidth	Use Case
Dedicated Interconnect	10-200 Gbps	Large, predictable transfers
Partner Interconnect	50 Mbps-50 Gbps	Medium bandwidth
HA VPN	1.4-3 Gbps/tunnel	Encrypted over internet

Cloud NAT

- Outbound-only internet access for VMs **without** external IPs
- No proxy VMs needed; software-defined; regional resource

Cloud CDN

- Works with External Application LB only
- Caches static content at Google edge PoPs
- Supports signed URLs/cookies

EXAM TRAPS & GOTCHAS

- :warning: Cloud Storage: ALL classes have **millisecond latency** – do NOT confuse with AWS Glacier
- :warning: Spanner is expensive (~\$0.90/node/hr) – don't pick it for small apps; Cloud SQL is cheaper

- :warning: Bigtable is NOT cost-effective for <1 TB of data
- :warning: Firestore Native mode = real-time + offline; Datastore mode = server-side only – **cannot switch modes after creation**
- :warning: Spot VMs: no SLA, 30s warning – never use for production serving or HA databases
- :warning: Sustained Use Discounts are **automatic** (no commitment); CUDs require commitment
- :warning: Custom machine types cost **more per unit** than predefined equivalents
- :warning: Cloud Functions Gen 1: 1 concurrent request/instance, 9 min timeout; Gen 2 fixes both
- :warning: Standard network tier does **NOT** support global load balancing
- :warning: Regional PD costs ~2x zonal PD – only use when HA justifies the cost
- :warning: Cloud Run scales to zero = zero cost when idle; GKE does NOT scale to zero (cluster fee always running)
- :warning: BigQuery on-demand: first 1 TB/month scanned is free; storage auto-drops to long-term rate after 90 days
- :warning: GKE Standard charges \$0.10/hr cluster management fee; Autopilot has no cluster fee (pay per pod)
- :warning: External Passthrough Network LB is the ONLY option for UDP traffic
- :warning: Cloud SQL max = 64 TB; if you need more, consider AlloyDB, Spanner, or Bigtable
- :warning: Sole-tenant nodes: for compliance (BYOL licensing, physical isolation) – not for performance
- :warning: Region choice factors: latency, compliance/data residency, pricing (us-central1 cheapest), service availability